ECCC Movebank Attribute Dictionary

For GPS archival & base station, GSM, and satellite trackers

Updated: 21 October 2026

This document defines the attributes (fields) to include in ECCC Movebank studies in which animals are tracked using archival or base-station GPS devices. It will help ensure consistency in the use of field names, units, and categorical variable definitions. Adherence to this system will enable users to programmatically access data from multiple projects and use them together in analyses without needing to clean and standardize data on a case-by-case basis.

This dictionary is largely based on the Movebank Attribute Dictionary (Max Planck Institute of Animal Behaviour, 2024) but only includes the Movebank data fields that we anticipate being used by ECCC. Fields required for Ecotone, Pathtrack, and TechnoSmArt GPS devices are currently included. Fields can be added to this dictionary as new tracking technologies are adopted. Please reach out to the generic emails for WRD Tracking East or West to get your device type added! We also suggest custom fields to add to your study that are not defined within the Movebank database.

Definitions are organized by data type: Study Attributes, Reference Attributes (Animal Attributes, Deployment Attributes, Tag Attributes), and Event Attributes. Each of these sections includes subsections that indicate whether the fields listed are required, required if applicable, or optional for ECCC projects. If there is a 'required' or 'required, if applicable' field for which you do not have data (e.g. you are uploading historical data) please include this field in your data but leave the contents blank. Fields are presented in alphabetical order within each section. We recommend using the Navigation Pane.

Study Attributes

The following terms are used to describe individual studies in Movebank.

Required

citation: One or more citations for the study. Include citations for published papers or reports describing the data in the study, or datasets derived from the study. If the data have

not been published, an 'unpublished data' or 'in progress' citation should be used, listing authors and a title for the dataset.
Example: McIntyre CL, Douglas DC, Collopy MW (2008) Movements of golden eagles (Aquila chrysaetos) from interior Alaska during their first year of independence. The Auk 125(1): 214-224. https://doi.org/10.1525/auk.2008.125.1.214
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000068

contact person: The name and username of the Movebank account for the person who is the contact for the study.
Example: Rolandisimo (Roland Kays)
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000072

grants used: A list or description of grants and other funding sources for the research.
Example: This project was funded by an NSF grant to RK.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000123

main location lat: The latitude of the reference location for the study, used to generate the study marker shown on the Movebank map. This location should represent the deployment site, or the approximate centroid of deployment sites if they are variable. For small
Example: 42.718
Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000222, modified

main location long: The longitude of the reference location for the study, used to generate the study marker shown on the Movebank map. This location should represent the deployment site, or the approximate centroid of deployment sites if they are variable.
Example: -73.856
Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000223, modified

principal investigator: The current, or most recent, principal investigator (PI) or lead researcher for the study. Reported either as the PI name or the name and username of the associated Movebank account. Additional PIs can be detailed in the study README file.
Example: Rolandisimo (Roland Kays)
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000164, modified

study ID: A unique identifier for the study in Movebank that is created automatically by the database.
Example: 2911040
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000172

study name: The name of the study in Movebank, following format 'SpeciesName BasicTagType – Organization – StudySite, Region'
Example: Leach's storm-petrel GPS - ECCC - Baccalieu Is., Atlantic Canada
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000173, modified

study summary: A description of the study objectives, methods, and results. Identify any permits and ethical approvals used during the completion of this work.
Example: Determine breeding season space use and foraging behaviour of Atlantic puffin in Witless Bay, Newfoundland, Canada. Tracking was done using GPS tags with a TDR sensor and base-station connectivity. Migratory birds permit 10559. Methods approved by the ECCC Wildlife Eastern Animal Care Committee, project 25GR01.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000176, modified

study type: Type of study. Standard studies are 'research' studies. (The alternative is a 'test' study which is automatically deleted in 90 days).
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000272

Required, if applicable

acknowledgements: Acknowledgements for the study. For example, people or institutions that provided logistics support or site access, and names of field assistants or collaborators.
Example: We thank Carrie Smith and John Johnson for help in the field and the Delaware Parks Association for access to our research site and logistical support.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000010

license terms: For use where license type is CUSTOM. Terms of use for the data in the study, provided by the study owner. Also see 'license type' for use of Creative Commons licenses. In addition to terms listed, the General Movebank Terms of Use and User Agreement apply for all studies.
Example: Data can be viewed publicly and used for educational purposes, but cannot be shared or used in publications without prior written permission from principal investigator.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000138

license type: The license that currently applies to data that are publicly downloadable from the study. Chosen by the study owner. If CUSTOM is selected, details may be available in 'license terms'. Allowed values are
CC_BY = Creative Commons Attribution (BY) License;
CC_BY_NC = Creative Commons Attribution-NonCommercial (BY-NC) License;
CC_0 = Creative Commons Universal (CC0) License;

CUSTOM = Custom terms of use defined by the study owner.
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000257

principal investigator address: Address, affiliation or other contact information for the principal investigator (PI) or lead researcher for the study. Used when the PI does not have a Movebank account.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000234

principal investigator email: Email address for the principal investigator (PI) or lead researcher for the study. Used when the PI does not have a Movebank account.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000235

Reference Attributes

Animal Attributes

These terms are used to describe individual animals. See 'Deployment Attributes' for information about animals that varies with time (e.g. conditions at capture and recovery).

Required

animal-id: An individual identifier for the animal, provided by the data owner. The recommended value is the band number, without a hyphen. If there is a chance there could be more than one deployment of the same tag-id on same animal during the study, the animal-id must be modified to ensure a unique value for each deployment (e.g. add deployment date, see Example 2). If band number is unknown, use format 'tag-id'-'spcd'-'study-site'-'deployYear' (Example 3). Deployment date can be added after tag-id if required to result in unique values. Values are unique within the study. If the data owner does not provide an animal-id, an internal Movebank animal identifier is sometimes shown.
Example: 246100765
Example 2: 246100765-20160701
Example 3: URI11-ATPU-GullIs-2016
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000016, modified

animal-sex: The sex of the animal. Allowed values are
m = male;
f = female;
u = unknown.

Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000023

animal-taxon: The scientific name of the taxon on which the tag was deployed, as defined by the Integrated Taxonomic Information System (ITIS). If the species name cannot be provided, this should be the lowest level taxonomic rank that can be determined and that is used in the ITIS taxonomy. Additional information can be provided using the term 'taxon detail'.
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000024http://vocab.nerc.ac.uk/collection/MVB/current/MVB000024

ring-id: Band number, without hyphen. Enter color bands and other markings in 'animal marker ID' if present.
Example: 246100765
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000022, modified

Optional

animal-mates: Band number, without hyphen, of the animal's mate in the year of deployment. This is a mandatory field ONLY if both members of the pair were tracked simultaneously. If the tracking period spans multiple years, include the year of the pairing as shown in Example 2 and separate years using a comma (example indicates both members of the pair were tracked starting in the 2024 breeding season, but that the tracked bird was observed with a different unbanded mate in 2025).
Example: 246100765
Example 2: 246100765-2024, unk-2025
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000342, modified

animal-marker-id: A description of unique identifiers or markings on the animal, for example color bands or ear tags. Bird ring or band numbers should be stored in 'animal-ring-id' (not here).
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000341

animal-taxon-detail: A more specific name and/or reference for the taxon name provided by 'animal taxon'. This can be used, for example, to specify a subspecies or a taxon not supported by the ITIS.
Example: Calonectris diomedea borealis (Cory, 1881)
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000025

Deployment Attributes

These terms are used to describe individual deployments of tags on animals.

Required

animal-life-stage: The age class or life stage of the animal at the beginning of the deployment. Can be years or months of age or terms such as 'adult', 'subadult' and 'juvenile'. Best practice is to define units in the values if needed (e.g. '2 years').
Example: adult
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000018

animal-mass: The mass of the animal at the beginning of the deployment. Measurement precision should be consistent for all birds in the study.
Example: 500
Units: grams
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000019

animal-reproductive-condition: The reproductive condition of the animal at the beginning of the deployment. Values are chosen from a controlled list. *Also see section *Custom fields* for reproductive condition at retrieval*.
incubation = bird has an egg;
chick rearing = bird has a chick;
breeding = bird has a brood patch, but nest contents are unknown;
nonbreeding = bird does not have a brood patch (if captured during the breeding season) or bird was captured outside the breeding season;
unknown = no other value applies (e.g. deploying late in the breeding season and can't tell if the brood patch has re-feathered or was never developed).
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000021, modified

attachment-body-part: The location of the tag on the animal. In some cases, this can also be indicated by 'attachment type'. If the tag is not attached on the mid-line of the animal, specify the side of placement to assist interpretation of sensor data.
Example: left leg
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000395, modified

attachment-type: The primary (strongest) attachment technique used to affix the tag to the animal. For example, a tag attached to a bird's feathers using TESA tape and reinforced using glue and cable ties would be assigned 'tape'. The details of the attachment method can be provided in 'attachment comments'. Values are chosen from a controlled list:
backpack-harness = The tag is attached to the animal using a backpack-style harness;

collar = The tag is attached by a collar around the animal's neck;
ear-tag = The tag is attached to the animal's ear;
fin mount = The tag is attached to the animal's fin;
glue = The tag is attached to the animal using glue;
harness = The tag is attached to the animal using a harness;
implant = The tag is placed under the skin of the animal or in the abdominal cavity [if selected, clarify placement details in field ‘attachment comments’];
leg-band = The tag is attached as a leg band or ring;
leg-loop-harness = The tag is attached to the animal using a leg-loop-style harness;
none = No tag was attached, e.g., for observations using natural markings;
other = The tag is attached using another method [if selected, describe in field ‘attachment comments’];
subcutaneous-anchor = The tag is attached using one or more anchors attached underneath the animal's skin;
suction-cup = The tag is attached using one or more suction cups;
sutures = The tag is attached by one or more sutures;
tape = The tag is attached to the animal using tape.
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000052

capture-method: A description of the method or methods used to bait, trap or immobilize the animal in order to attach the tag.
Example: One-way trap at burrow entrance
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000397

deploy-off-timestamp: The timestamp when the tag deployment ended. Data records recorded after this day and time are not associated with the animal related to the deployment. Values are typically defined by the data owner and should indicate the timestamp of recapture or the timestamp at which the tag stopped collecting reliable data (but was still deployed or had fallen off the animal). Further information can be provided in 'deployment end type' and 'deployment end comments'. This field can be left blank if you are uploading data from a deployment which has not yet ended.
Example: 2009-10-01 12:00:00.000
Format: yyyy-MM-dd HH:mm:ss.SSS
Units: UTC
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000077, modified

deploy-on-latitude: The geographic latitude of the location where the animal was released. Intended primarily for cases in which the animal release location has higher

accuracy than that derived from sensor data.
Example: 27.3516
Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000078

deploy-on-longitude: The geographic longitude of the location where the animal was released. Intended primarily for cases in which the animal release location has higher accuracy than that derived from sensor data.
Example: -97.3321
Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000079

deploy-on-timestamp: The timestamp when the tag deployment started. Data records recorded before this day and time are not associated with the animal related to the deployment. Values are typically defined by the data owner, and in some cases are created automatically during data import.
Example: 2008-08-30 18:00:00
Format: yyyy-MM-dd HH:mm:ss
Units: UTC
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000081, modified

deployment-id: A unique identifier for the deployment of a tag on animal, provided by the data owner. Formatted as 'tag ID' and 'animal ID', separated by a hyphen.
Example: CD456-246100765
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000085, modified

duty-cycle: Remarks associated with the duty cycle of a tag during the deployment, describing the times it is on/off and the frequency at which it transmits or records data. Units and time zones should be defined in the remarks. This field is limited to 100 characters. Additional details should be entered in field 'tag-settings' as required. If using the tag-settings column, please include 'see tag-settings' within the duty-cycle field.
Example: GPS@10min, off from 0:00-10:00 UTC (8:00-18:00 ADT)
Example2: Duty cycle varies by year. See tag-settings.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000086

study-site: A location name such as the deployment site, study site, or colony.
Example: Baccalieu Island
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000175

tag-mass-total: The total mass of the tag and related hardware, such as housing and harness, during the deployment. The mass of the tag itself should be defined in 'tag mass'

(Tag Attributes section) for clarity. This field may be left blank ONLY if the data being archived are historical and the total tag mass cannot be determined.
Units: grams
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000409, modified

tag-readout-method: The way the data are received from the tag. Values are chosen from a controlled list:
ISS = Data are transferred via the International Space Station;
LPWAN = Data are transferred through a low-power wide-area network, such as LoRa or Sigfox;
multiple = Data are acquired using multiple methods;
none = Data are obtained without use of an animal-borne tag, such as by observing a unique marking;
other-wireless = Data are transferred via another form of wireless data transfer, such as a VHF transmitter/receiver;
phone-network = Data are transferred via a phone network, such as GSM or AMPS;
satellite = Data are transferred via satellite;
tag-retrieval = The tag must be physically retrieved in order to obtain the data;
telemetry-network = Data are obtained through a radio or acoustic telemetry network;
Wi-Fi/Bluetooth = Data are transferred via a local Wi-Fi or Bluetooth system.
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000188

Required, if applicable

attachment-comments: Additional comments about how the tag is attached to the animal. Required if the value provided for 'attachment type' could have multiple interpretations (e.g. implant), and if additional information is required to interpret acceleration or other sensor data collected by the tag. Materials used may also be included for comparison of tag retention rates etc.
Example (implant): Tag inserted into coelomic cavity.
Example (back-mounted): Top of tag located between scapulae. Four ½ inch strips of TESA tape, each around a group of feathers comprised of several tracts plus the device. Ultragel to seal tape seams, one stainless steel tooth ziptie around entire assembly.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000396, modified

deploy-off-latitude: The geographic latitude of the location where the deployment ended. Intended primarily for cases in which the recapture, tag retrieval or other final location has higher accuracy than that derived from sensor data.
Example: -38.6866

Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000074

deploy-off-longitude: The geographic longitude of the location where the deployment ended. Intended primarily for cases in which the recapture, tag retrieval or other final location has higher accuracy than that derived from sensor data.
Example: 146.3104
Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000075

tag-calibration: A description of how sensors (magnetometer, temperature sensor) on the tag were calibrated. Reference to a calibration file can also be provided.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000408

tag-settings: A description of tag and sensor settings during the deployment. Reference to a settings file can also be provided. This field is required if any settings are used beyond those reported in 'duty-cycle'.
Example: Tag programmed not to attempt a fix when wet (salt water) and to stop attempting a fix when dry after 2min.
Example2: GLS tag programmed to record in Mode 6. Light sensor samples at full range (1-74,000lux) every minute, records as max lux every 5 min. Wet-dry sensor samples conductivity every 30 sec and records the number of wet counts every 10 min.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000411

Optional

See section *Custom Fields* for more optional deployment fields.

capture-handling-time: The time elapsed between capture and release of the animal for the purpose of attaching the tag. Can also be calculated as the difference between 'deploy-on timestamp' and 'capture timestamp'.
Example: 552
Units: seconds
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000349

deploy-off-person: The name of the person/people who removed the tag from the animal and ended the deployment. It is preferable to list all people on the retrieval team if person is unknown rather than leaving the field blank.
Example: J. Smith
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000076

deploy-on-person: The name of the person/people who attached the tag to the animal and began the deployment. It is preferable to list all people on the retrieval team if person is unknown rather than leaving the field blank.
Example: G. Smith
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000080

deployment-comments: Additional information about the tag deployment that is not described by other reference data terms.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000082

deployment-end-comments: Additional information about the end of a tag deployment. Also see 'deployment end type' and 'tag failure comments'.
Example: Only one suture still attached.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000083, modified

deployment-end-type: A categorical classification describing the end of the tag deployment on the animal. Best practice is to clarify how the 'deploy-off timestamp', if present, was chosen. Values are chosen from a controlled list. The most commonly used values are highlighted in green.
analysis-end = the end time represents the end of the period of interest;
captured = The tag remained on the animal but the animal was captured or confined;
dead = The deployment ended with the death of the animal that was carrying the tag;
dead/fall-off = The tag stopped moving, and it is not possible to determine whether it is due to death of the animal or unscheduled tag detachment;
equipment-failure = The tag stopped working [clear evidence that the battery or sensor was failing before it stopped collecting data];
fall-off = The attachment of the tag to the animal failed, and it fell of accidentally [animal was recaptured or observed without the tag];
other = other;
released = The tag remained on the animal but the animal was released from captivity or confinement;
removal = The tag was purposefully removed from the animal [while the tag was still collecting good quality data];
scheduled-detachment = The tag was programmed to detach from the animal;
transmission-end = The tag stopped transmitting usable data;
unknown = The cause of the end of data availability or transmission is unknown.
Note: For tags with remote data download, 'unknown' should be used if the animal was not recaptured or observed again (unclear if the tag stopped working, fell off, or the animal died). An exception would be if the last tag transmission clearly indicated that the tag was

failing (nonsensical data) in which case use 'equipment-failure' and enter a 'deploy-off-timestamp' that coincides with the last reliable datapoint.
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000084, modified

Custom Fields

Movebank only provides one field each for any measurements or tissue samples that were taken at the time of tag deployment or tag recovery. The 'best practice' is to nest your own subfields (e.g. deploy off HPAI swab, deploy off fecal sample) into these columns so they can be downloaded and recreated programmatically using a 'key:value encoding schema' such as JSON. We have found this to be unnecessarily complicated (and frustrating!) to implement, especially given how these data are likely to be used. Instead, we suggest including these fields as **unmapped columns** in the reference data file you upload to Movebank. Although these fields will not be integrated into Movebank's database, they will be archived in a copy of your original reference data file within the Attachments section of your project, which can be manually downloaded.

While we could include any custom fields we want, we should agree on standardized field names and units for the fields we think ECCC will use most often. If you have suggested modifications or additions to the list below, please contact the generic emails for WRD Tracking East or West.

Deployment Context

deploy-off-animal-reproductive-condition: The reproductive condition of the animal at the end of the deployment. Movebank already has a supported field 'animal reproductive condition' that is specific to the beginning of the deployment, so do not use 'deploy on animal reproductive condition'. This field helps flag whether the bird changed breeding status during the deployment. Values are chosen from a controlled list:
incubation = bird has an egg;
chick rearing = bird has a chick;
breeding = bird has a brood patch, but nest contents are unknown;
nonbreeding = bird does not have a brood patch (if captured during the breeding season) or bird was captured outside the breeding season;
unknown = condition unknown at the time deployment ends (e.g. tag not retrieved or falls off and burrow was not monitored during that time).

Measurements

*For all field names below, please add **'deploy-on'** or **'deploy-off'** at the beginning as appropriate. There is one exception: Movebank already has a field 'animal mass' specific to*

the animal mass at the time of deployment (see Deployment Attributes: Required), so do not use 'deploy on animal mass'.

deploy-off-animal-mass: The mass of the animal at the end of the deployment. Measurement precision should be consistent for all birds in the study. Movebank already has a supported field 'animal mass' that is specific to the beginning of the deployment. Do not use 'deploy on animal mass'.
Example: 435
Units: grams

-culmen-length-exposed: The culmen length measured from where the feathers begin at the base of the bill (or base of the nasal tube) to the tip of the bill. Measured to the nearest 0.1 mm.
Example: 46.5
Units: millimeters

-bill-depth-base: The bill depth measured where the feathers begin at the base of the bill (puffins) or at the furthest reach of feathering on the upper mandible (terns), perpendicular to the long axis of the bill. Measured to the nearest 0.1 mm.
Example: 37.1
Units: millimeters

-bill-depth-gonys: The bill depth measured at the deepest part of the gonys, perpendicular to the long axis of the bill. Measured to the nearest 0.1 mm.
Example: 37.1
Units: millimeters

-bill-depth-nares: The bill depth measured at the front of nares, perpendicular to the long axis of the bill. This field is also applicable to Procellariids (measured right in front of the tube) and to rhinoceros auklets (measured right in front of the horn). Measured to the nearest 0.1 mm.
Example: 37.1
Units: millimeters

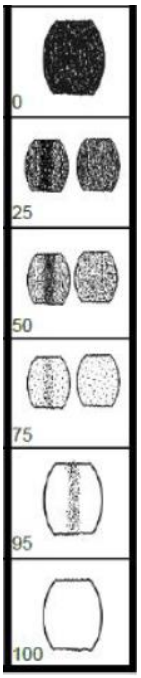
-bill-grooves: The number of bill grooves in an Alcid bill. While this is a simple count, increments of 0.5 are allowed to indicate partial grooves.
Example: 1.5

-bill-width-base: The bill width measured where the feathers begin at the base of the bill. For puffins, this measurement occurs at the rosettes. Measured to the nearest 0.1 mm.
Example:
Units: millimeters

-horn-height-total: The measurement from the top of the horn to the gonys. Used only for rhinoceros auklet. Measured to the nearest 0.1 mm.
Example:
Units: millimeters

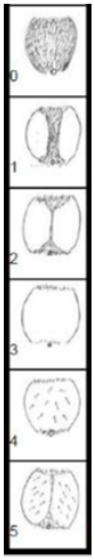
- moult-percent-skin-broodpatch: Moult stage of brood patch, described as percent bare skin (fully feathered = 0%, bare skin = 100%; Figure 1).
Example: 50

Figure 1. Guide to describing brood patch moult as percent bare skin.



- moult-score-broodpatch: Moult score for the brood patch. Scores may be assigned from 1 to 5 as described in Figure 2, or may simply be classified as present (“P”) or absent (“0”).
Example 1: 3
Example 2: P

Figure 2. Guide to scoring brood match moult.



0 – absent

1 – starting

2 – well defined, breast muscle and gut still visible through the skin

3 – the skin of the belly is opaque, thickened or engorged, veined and red with broad, undulating wrinkles

4 – skin of the belly has thin wrinkles, engorgement of the skin gone, but the skin is still stretched, and the breast muscle and gut are visible again through the skin

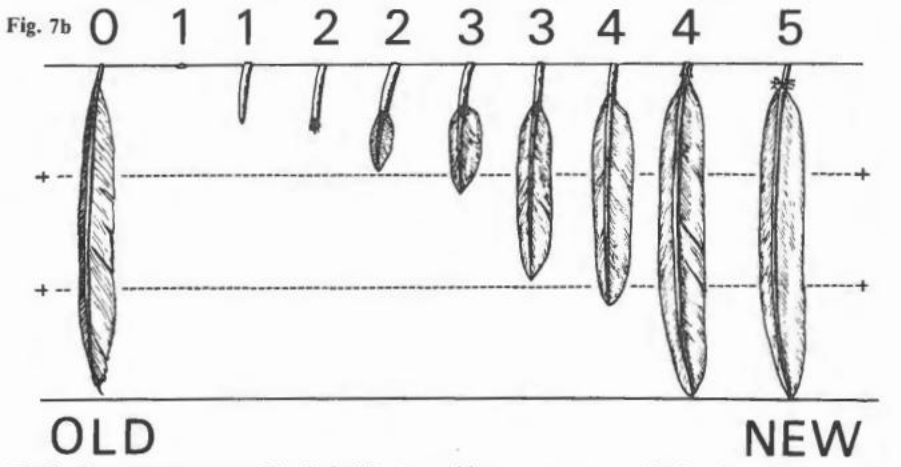
5 – feathering over

-moult-score-primaries: Moult scores for primary feathers. Indicate which wing was scored with "L:" or "R:", then list the moult scores in order from innermost feather to outermost feather (P1 to Px). Each feather should receive a score from 1 to 5, according to the British Trust for Ornithology (BTO) standard (Figure 3, below).
Example: "L:0000000000, R:0000000000"

-moult-score-retrices: Moult scores for retrix feathers. Indicate which side of the tail was scored with "L:" or "R:", then list the moult scores in order from innermost feather to outermost feather (R1 to Rx). Each feather should receive a score from 1 to 5, according to the British Trust for Ornithology (BTO) standard (Figure 3, below).
Example: "L:000000, R:000000"

-moult-score-secondaries: Moult scores for secondary feathers. Indicate which wing was scored with "L:" or "R:", then list the moult scores in order from innermost feather to outermost feather (S1 to Sx). Each feather should receive a score from 1 to 5, according to the British Trust for Ornithology (BTO) standard (Figure 3, below).
Example: "L:0000000000000, R:0000000000000"

Figure 3. BTO standard for scores of moulting feathers:



The feather scoring system. The dashed horizontal lines represent one third and two thirds growth. The numerical scoring system is:

0 Old feather remaining.
1 Old feather missing or new feather completely in pin.
2 New feather just *emerging from the sheath up to one third grown.*
3 New feather between one and two thirds grown.
4 New feather more than two thirds grown and with remains of waxy sheath at its base.
5 New feather fully developed with no trace of waxy sheath remaining at base.

-tarsus-diagonal: The length of the tarsus as measured from the notch in the heel to the end of the tarsus bone exposed by bending the bird's foot downward away from its head. Also known as tarsus length from wrist. Measured to the nearest 0.01 mm.
Example: 27.6
Units: millimeters

-tarsus-straight: The length of the tarsus as measured from the notch in the heel to the end of the tarsus bone exposed by bending the bird's foot upward toward its head. Also known as tarsus length from palm. Measured to the nearest 0.01 mm.
Example: 27.6
Units: millimeters

-total-head-length: The measurement from the back of the head to the tip of the bill. Measured to the nearest 0.01 mm.
Example:
Units: millimeters

-wing-flat: The wing chord measured to the nearest millimeter from the wrist (carpel joint) to the tip of the longest primary with the feathers flattened. Wing held bent at a 90-degree angle. Measured to the nearest 1mm.
Example: 173
Units: millimeters

-wing-natural: The wing chord measured to the nearest millimeter from the wrist (carpel joint) to the tip of the longest primary with the wing held so the feathers retain their natural curvature. Wing held bent at a 90-degree angle. Measured to the nearest 1mm.
Example: 173
Units: millimeters

Sampling

*For all field names below, please add **'deploy-on'** or **'deploy-off'** at the beginning as appropriate.*

-blood-card-sex: Whether a blood card was collected for genetic sexing.
Options: yes, no

-blood-card-cort: Whether a blood card was collected for corticosterone assay at the end of deployment.
Options: yes, no

-blood-collection-method: Location of the bleeding site and optionally how the blood was collected.
Example: prick at brachial vein, collect by capillary tube

-blood-nobuto: Whether a Nobuto strip was collected. If the strip was collected but was not fully saturated, use 'partial'.
Options: yes, partial, no

-blood-storage-method: Description of how the blood was handled and stored after collection. If it is different from final storage, note how the sample was stored in the field and for how long. If blood was stored in media, provide specifics (e.g. 90% ethanol, type of buffer solution). If you collected more than one type of blood sample, please make sure it is clear which you are describing. This is a good place to flag if there may be issues with a particular sample, such as prolonged heat exposure.
Example: Blood stored 2h in field cooler with ice, then plasma and red blood cells separated by centrifugation and frozen in nitrogen dewar. Final storage in -80 freezer.

-blood-volume: If blood was collected by capillary tube or syringe, what volume of blood was collected.
Example: 1.5
Units: milliliters

-blood-volume-use: If blood was collected by capillary tube or syringe, what is the intended use of the sample (optionally, who was the sample for).
Example: Trace elements analysis (Dr. Alex Bond), HPAI (Dr. Andrew Lang)

-feather-sample: Type(s) and number(s) of feather sampled. Well-recognized feather abbreviations may be used (e.g. P = primary, S = secondary, R = retrix) but others should be written in full. For wing feathers, the number sampled can be implied by including R or L to indicate which wing was sampled.
Example: P1R, S4L, facial plume - 3

-swab-eDNA: The sample ID for eDNA fecal swab collected.
Example: GI-AP-2024-17

-swab-HPAI: The sample ID for HPAI oral/cloacal swab collected.
Example: GI-AP-2024-17

-ticks-method: The method used to check for ticks.
Example: Timed full body check for 60 seconds
Example2: Opportunistic

-ticks-number: The number of ticks found.
Example: 2

-ticks-sample: The sample ID for ticks if collected.
Example: GI-AP-2024-17

Tag Attributes

These terms are used to describe individual tags. See 'Deployment Attributes' for information about tags that can vary between deployments.

Required

tag-id: An identifier for the tag, provided by the data owner. Values are unique within the study.
Example: CD552
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000181

tag-manufacturer-name: The company or person that produced the tag.
Example: Pathtrack Ltd
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000183

tag-mass: The mass of the tag. Can be used with 'tag mass total' (Deployment Attributes section) to define the mass of the tag separately from that of the tag with mounting materials.
Example: 24

Units: grams
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000184

tag-model: The model of the tag.
Example: NanoFix-GEO mini
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000185

Optional

tag-comments: Additional information about the tag that is not described by other reference data terms.
Example: Pathtrack GPS manufactured in 2020 with poor performance, refurbished and battery replaced in 2021
Example 2: Custom-made leg-mounted Pathtrack GPS with a special altitude sensor
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000179

tag-failure-comments: Comments about tag malfunction or failure.
Example: tag stopped transmitting 2023/12/08, was not found
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000180

tag-production-date: The approximate date the tag was produced (can be a year, month, or day).
Example: March 2011
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000187

tag-serial-no: The serial number of the tag.
Example: MN93-33243
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000189

Event Attributes

These terms are used to describe individual locations or other sensor measurements (events). This version of the Attribute Dictionary covers fields for archival or base station download GPS data and associated temperature-depth (TDR) data. If you have suggested modifications or additions to the list below, please contact the generic emails for WRD Tracking East or West.

All Event Types

Required

tag-id: See Section 'Tag Attributes' for details. This attribute is required in both the Reference dataset and the Event-level datasets. It is used along with the Event attribute 'timestamp' (this section) to associate event-level data with the appropriate Reference data and identify deployments using the 'deploy-on-timestamp' and 'deploy-off-timestamp'.

event-id: An identifier for the set of values associated with each event, i.e. sensor measurement. A unique event ID is **automatically assigned** to every time-location or other time-measurement record in Movebank. If multiple measurements are included within a single row of a data file, they will share an event ID. If users import the same sensor measurement to Movebank multiple times, a separate event ID will be assigned to each.
Example: 14328243575
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000103

sensor-type: The type of sensor with which data were collected. All sensors are associated with a tag id, and tags can contain multiple sensor types. Each event record in Movebank is assigned one sensor type. If values from multiple sensors are reported in a single event, the primary sensor is used. Values are chosen from a controlled list:
acceleration = The sensor collects acceleration data;
accessory-measurements = The sensor collects accessory measurements, such as battery voltage;
acoustic-telemetry = The sensor transmits an acoustic signal that is detected by receivers to determine location;
argos-doppler-shift = The sensor location is estimated by Argos using Doppler shift;
atlas-geolocation = The sensor location is estimated by an ATLAS system;
barometer = The sensor records air or water pressure;
bird-ring = The animal is identified by a band or ring that has a unique identifier;
derived = Information is derived from other sensor data;
geolocation-api = The location is determined based on detection by nearby cell towers or other wireless network connectors using the World Wide Web Consortium Geolocation API Recommendation;
gnss = The sensor uses one or more global navigation satellite systems to determine location;
gps = The sensor uses GPS to determine location;
gyroscope = The sensor records angular velocity;
heart-rate = The sensor records or is used to calculate heart rate;

magnetometer = The sensor records the magnetic field;
natural-mark = The animal is identified by a unique natural marking;
orientation = Quaternion components describing the orientation of the tag are derived from accelerometer and gyroscope measurements;
proximity = The sensor identifies proximity to other tags;
radio-transmitter = The sensor transmits a radio signal that is detected by receivers to determine location;
sigfox-geolocation = The sensor location is determined by Sigfox using the received signal strength indicator;
solar-geolocator = The sensor collects light levels, which are used to determine position (for processed locations);
solar-geolocator-raw = The sensor collects light levels, which are used to determine position (for raw light-level measurements);
solar-geolocator-twilight = The sensor collects light levels, which are used to determine position (for twilights calculated from light-level measurements);
tdr = The sensor is a time-depth recorder.
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000170

timestamp: The date and time in UTC corresponding to a sensor measurement or an estimate derived from sensor measurements.
Example: 2008-08-14 18:31:00.000
Format: yyyy-MM-dd HH:mm:ss.SSS
Units: UTC
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000200, modified

Optional

breed-stage: The breeding stage of the animal. This is included here instead of in 'Animal Attributes' because breeding status can change over the course of a deployment.
Example: chick-rearing
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000431

comments: Additional information about events that is not described by other event data terms. Movebank field 'outlier-comments' is no longer functional. Use this column to describe or reference the methods used to define outliers in 'import marked outlier' and/or 'manually marked outlier'. If outliers were identified prior to import ('import marked outlier'), this field can be used to describe why the data were considered outliers. This field can also be used to indicate that values were modified prior to import, and why.

Example: Pathtrack location 0,0 & last good position >5km from colony
Example 2: NOT filtered - reassigned to deployment long,lat; Pathtrack location 0,0 & last good position <=5km from colony
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000069, modified

import-marked-outlier: Identifies events as outliers. Outliers have the value TRUE. Typically used to import a record of outliers that were identified by the data provider or owner with automated methods outside of Movebank. Information about how outliers were defined can be provided in 'outlier comments'. The script used to identify outliers should also be uploaded to the 'Attachments' section of the Movebank project.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000133, modified

manually-marked-outlier: Identifies events flagged manually as outliers, typically using the Event Editor in Movebank. Outliers have the value TRUE. Information about how outliers were defined can be provided in 'outlier comments'.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000156, modified

GPS

Required

gps-fix-type-raw: The type of GPS fix as provided by the tag. Common values are
1D = no fix
2D = altitude typically not valid
3D = altitude typically valid
Example: 3D
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000117

gps-satellite-count: The number of GPS satellites used to estimate the location.
Example: 8
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000120

location-lat: The geographic latitude of the location as estimated by the sensor.
Example: -41.0982423
Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000145

location-long: The geographic longitude of the location as estimated by the sensor. Positive values are east of the Greenwich Meridian, negative values are west of it.
Example: -121.1761111
Units: decimal degrees, WGS84 reference system
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000146

tag-voltage: The voltage as reported by the tag, in millivolts.
Example: 2895
Units: mV
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000191

Required, if applicable

height-above-ellipsoid: The estimated height above the ellipsoid, typically estimated by the tag. [This is the type of altitude reported by Pathtrack tags. Other attribute field options include 'height above mean sea level', 'height above ground level', 'barometric height'.]
Example: 24.8
Units: meters
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000130

height-above-mean-sea-level: The estimated height of the tag above mean sea level, typically estimated by the tag. [This is the type of altitude reported by TechnoSmart tags. Other attribute field options include 'height above ellipsoid', 'height above ground level', 'barometric height'.]
Example: 34
Units: meters
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000131

gps-hdop: Horizontal dilution of precision provided by the GPS.
Example: 1.2
Units: unitless
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000438

gps-time-to-fix: The time required to obtain the GPS location fix.
Example: 36
Units: seconds
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000121

TechnoSmart-signal-quality: The strength of satellite reception. Ranges from 0 to 500. Higher values indicate better reception.
Example: 230
Units: dB-Hz
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000193

Optional

ground-speed: The estimated ground speed provided by the sensor or calculated between consecutive locations. *Please double check what units your tracking device reports and convert as necessary. Ecotone reports speed in knots!*
Example: 7.22
Units: m/s
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000124, modified

TDR

Required

barometric-pressure: The barometric air or water pressure.
Example: 32536.0
Units: mbar (hPa)
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000055

external-temperature: The temperature measured by the tag (different from ambient temperature or internal body temperature of the animal).
Example: 32.1
Units: degrees Celsius
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000104

Required, if applicable

depth: The device depth below the water surface.
Example: 16.4
Units: meters
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000477

dive-duration: The duration of a dive event beginning at the timestamp.
Example: 45
Units: seconds
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000399

TechnoSmart-activity: An indication of whether an animal is active. Possible values are 'active' and 'inactive'. The threshold used to define activity is dependent on how the tag is configured by the user.
Format: controlled list
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000192

wet: Whether the sensor is wet, as detected by a conductivity sensor. Allowed values are TRUE or FALSE.
URI: http://vocab.nerc.ac.uk/collection/MVB/current/MVB000299

Other Accessory Data

Please consult Movebank's Attribute Dictionary for field names and units applicable to other accessory data types. We welcome recommendations to add accessory data types to this dictionary, including ECCC required, required if applicable, and optional fields. Please email your suggestions to WRD Tracking East and West to ensure a response.

References

Max Planck Institute of Animal Behavior. 2026. Movebank Attribute Dictionary. British Oceanographic Data Centre, Natural Environment Research Council Vocabulary Server. http://vocab.nerc.ac.uk/collection/MVB/current accessed on 11 May 2026.